Daffodil Institute of IT (DIIT)

Department of Computer Science and Engineering

Project Proposal of Final Semester - 2026

Next-Generation AI-Driven Smart Education Management System

Enterprise Multi-Tenant SaaS Platform with Autonomous AI & Real-Time Voice Assistance

■ Project Contributors (Group Members)

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Session: 2020–2021 | **Dept:** CSE, DIIT
Focus: System Architecture, AI Pipeline & Forecasting

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Session: 2020–2021 | **Dept:** CSE, DIIT
Focus: Core ERP Backend, 32-Module RBAC & Auth Hub

■ Main Objectives

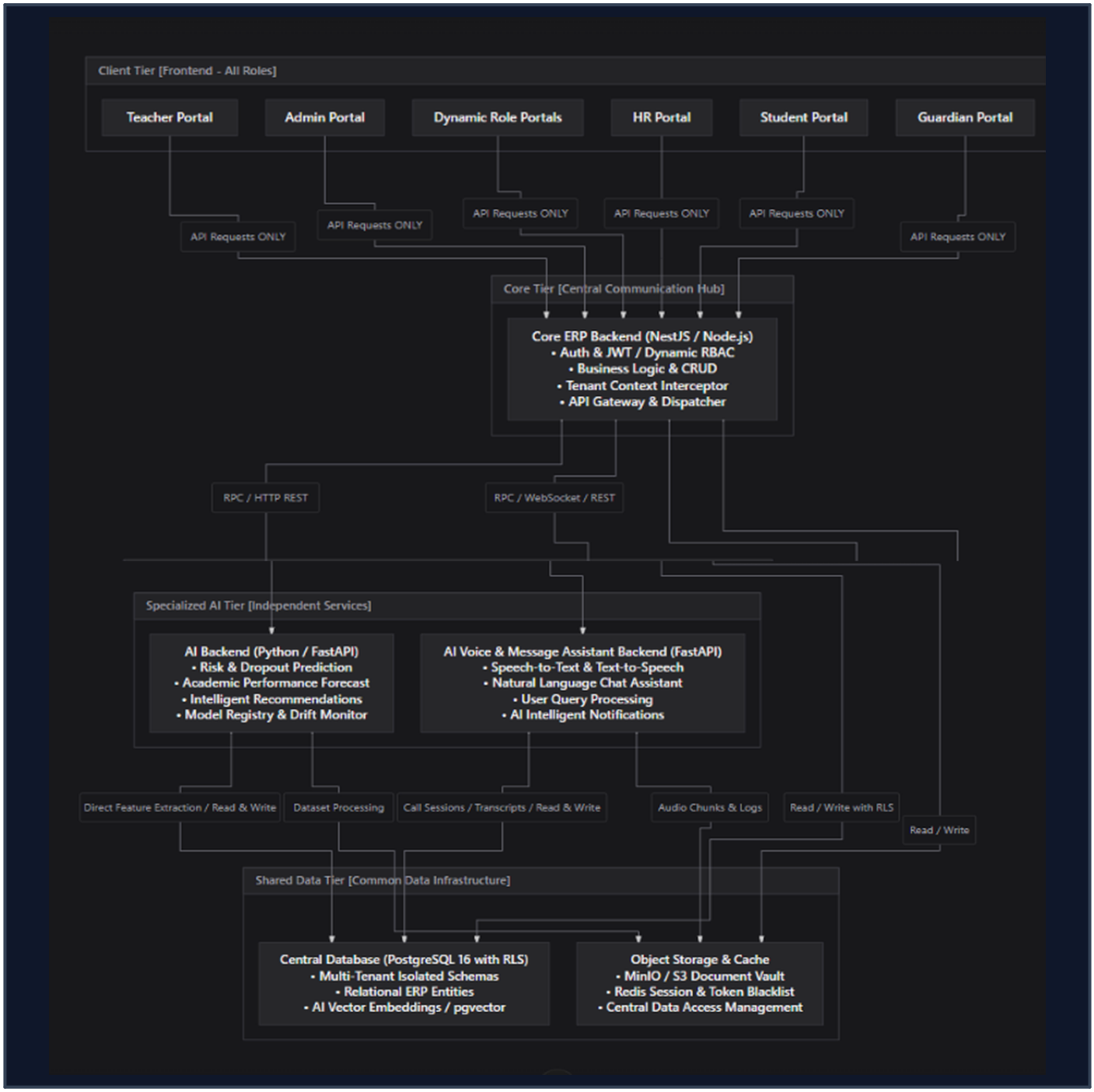
- **Real-Time Performance & Attendance Tracking:** Continuous monitoring of GPA trajectories, assignment workflows, and biometric attendance logs in real time.
- **AI-Driven Personalized Learning Paths:** Machine learning models identifying student learning gaps and recommending tailored pedagogical resources.
- **Automated Dual-Channel Parent Communication:** Real-time SMS, Email, and Push alerts notifying guardians of attendance anomalies, grades, and fee dues.
- **Faculty Insights & Dropout Risk Prediction:** Providing teachers with cohort failure risk alerts and data-driven teaching improvement strategies.

■ Key Features (Existing System Analysis)

Functional Cluster	Production Modules & System Capabilities
■ **Intelligent AI & Analytics**	• **AI Academic Risk Predictor:** Early dropout detection and GPA forecasting via ML regression/classification. • **AI Voice & Chatbot Assistant:** Independent conversational voice/text engine for inquiries & alerts. • **Timetable Conflict Detection:** Autonomous prevention of room, teacher, and schedule clashes.
■■ **Universal Security & RBAC**	• **Universal Credential Engine:** Auto-generates unique usernames and cryptographic passwords for all roles. • **Zero Plain-Text Policy:** bcrypt/Argon2id hashing with mandatory password change on first login. • **32-Module Dynamic RBAC Matrix:** Granular actions (create, read, update, delete, export, approve).
■ **Academics & Lifecycle**	• **Student & Faculty Management:** Comprehensive digital records, transcripts, and ID card barcode generation. • **Dual Admission Pipeline:** One-click admission simultaneously provisions student and guardian portal credentials. • **Examination & Grading:** Seat plans, dynamic grading policies, CGPA tabulation, and QR-verifiable PDF transcripts.
■ **Finance & Operations**	• **Fee & Invoicing Ledger:** Flexible fee heads, automated monthly billing, online payment gateway & ledgers. • **Operations Management:** Dedicated modules for Library Catalogs, Hostel Allocations, Fleet/Transport Routes.

■ Methodology & System Architecture

The system enforces a **Decoupled Tiered Microservices Architecture**. The Client Tier interacts exclusively with the Core ERP Central Communication Hub via secure RESTful APIs. The AI and Voice Assistant engines operate as autonomous microservices, directly accessing the PostgreSQL 16 Shared Data Infrastructure with Row-Level Security (RLS) isolation.



Tier	Enterprise Technology Stack & Responsibilities
Client Tier	**Next.js 16 App Router**, React 19, Tailwind CSS, Lucide Icons, Role Guards (Admin, Teacher, Student, Guardian Portals).
Core Hub	**Node.js / Express / TypeScript**: Central API Gateway, JWT Authentication, Dynamic RBAC, Multi-Tenant Interceptor.
AI Backend	**Python / FastAPI**: ML/DL models (Scikit-Learn, PyTorch) for student risk forecasting and grade predictive trajectory.
Voice Tier	**Python / FastAPI / WebSockets**: Speech-to-Text (STT), Text-to-Speech (TTS), and Natural Language Chat Query Processing.
Shared Data	**PostgreSQL 16** (RLS & pgvector), **Redis** (Session Cache & Token Blacklist), **MinIO / S3** (Encrypted Document Vault).

■ Research and Business Perspective

■ 5.1 Research Perspective & Scientific Novelty

• **Predictive Student Analytics:** Investigating supervised learning classifiers (Random Forest, Gradient Boosting, LSTMs) to identify academically at-risk students 6–8 weeks before final examinations based on attendance frequency, continuous assessment trends, and LMS engagement.
• **Multilingual Voice Interactivity in Education:** Exploring domain-adapted lightweight LLMs and automated speech engines to handle localized voice inquiries for parents and students with varying technical literacy.
• **Algorithmic Fairness & Model Governance:** Implementing drift monitoring, explainable AI (XAI), and zero-data-leakage architecture ensuring compliance with academic privacy regulations (GDPR / FERPA).

■ 5.2 Business & SaaS Perspective

• **Subscription-Based Multi-Tenant SaaS Model:** Tiered B2B subscription pricing (Basic, Standard, Enterprise) charged per-active-student annually, generating predictable, high-margin Annual Recurring Revenue (ARR).
• **Commercial Scalability & Market Demand:** High domestic and global demand across schools, colleges, and polytechnics seeking to replace fragmented legacy desktop tools with automated cloud ERPs.
• **Ecosystem Interoperability & White-Labeling:** Architecture ready for multi-currency payment gateways (bKash, Nagad, Stripe, SSLCommerz) and open LMS integrations (Moodle, Canvas, Google Classroom).
• **Ancillary Revenue Channels:** Additional monetization streams via premium AI predictive analytics add-ons, enterprise custom form builders, dedicated cloud backup vaults, and SMS/Voice communication gateway margins.

Formal Submission & Institutional Endorsement

Submitted By: MD. Maruf Ahmmed (Class ID: 210006) Sojib Chandra Roy (Class ID: 210005) Session: 2020–2021	**Submitted To:** Department of Computer Science and Engineering Daffodil Institute of IT (DIIT) Academic Year: 2026	**Project Status:** Final Semester Project Proposal Architecture & Backend Integrated Presentation Ready

Next-Generation AI-Driven Smart Education Management System — Designed and Engineered for Presentation at Daffodil Institute of IT (DIIT). All architectural diagrams and modular specifications verified.